

Introduction

Combining most of the topics we've just learned about, we'll turn our focus to the study of Groups. Groups are prevalent throughout many fields, where they are primarily used to describe symmetries. They appear in mathematics, physics, chemistry, engineering, and computer science. We will barely scratch the surface of Group Theory, but it will give you a taste of abstract algebra.

Learning Objectives

- Define a group and give simple examples of groups.
- Define an abelian group, and give examples of abelian and non-abelian groups.
- Define the symmetric group and dihedral groups. Prove basic facts about these groups.
- Prove simple properties of abstract groups.
- Define the order of a group and the order of an element. Be able to find the order of an element, and prove simple facts about order.
- Define a subgroup, and check whether a set is a subgroup or not. Prove simple facts about subgroups.
- Define cyclic groups, and prove simple facts about cyclic groups.

Groups – First Principles

Given a set G , a *binary operator on G* is a function $b : G \times G \rightarrow G$. Addition and multiplication are examples of binary operators on $\mathbb{R}$. While function notation would dictate that we write $b(g_1, g_2)$, when the operator is well-understood, we might write something like $g_1 + g_2$ (for addition) or $g_1 \cdot g_2$ for multiplication.

Definition 1

If G is a set and $\cdot$ is a binary operator, the pair $(G, \cdot)$ is said to be a *group* if it satisfies the following three properties:

- [Associativity] For every $a, b, c \in G$, $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.
- [Identity] There exists an element $e \in G$ such that $e \cdot a = a \cdot e = a$ for all $a \in G$.
- [Inverses] For every $a \in G$, there exists a $b \in G$ such that $a \cdot b = b \cdot a = e$. We often write the inverse of a as a^{-1} .

Instead of referring to the pair $(G, \cdot)$, we sometimes say that “ G is a group under the binary operator $\cdot$.” So for example, the reals $\mathbb{R}$ are a group under addition, as are the rationals $\mathbb{Q}$ and the integers $\mathbb{Z}$. However, the positive integers are not a group under addition, nor are the reals under multiplication (why?). As usual, we sometimes use concatenation to signify the group operation, writing $ab = a \cdot b$. For a positive integer n we may write n -fold product $a \cdot a \cdots a$ as a^n ,

and similarly a^{-n} for $(a^{-1})^n$. By convention, we take $a^0 = e$.

Exercise 2 Convince yourself that $(a^{-1})^n = (a^n)^{-1}$.

Example 3

Show that for any $n \in \mathbb{Z}^+$, $(\mathbb{Z}_n, +)$ is a group.

Solution. We need to prove the three identities. Let $n \in \mathbb{Z}^+$ be arbitrary and fix $a, b, c \in \mathbb{Z}_n$.

- [Associativity] Note that

$$\begin{aligned} [a]_n + ([b]_n + [c]_n) &= [a]_n + [b + c]_n = [a + (b + c)]_n \\ &= [(a + b) + c]_n && \text{addition is associative in } \mathbb{Z} \\ &= [a + b]_n + [c]_n = ([a]_n + [b]_n) + [c]_n. \end{aligned}$$

- [Identity] We claim that $[0]_n$ is the identity. Indeed, if $[a] \in \mathbb{Z}_n$ then $[a]_n + [0]_n = [a + 0]_n = [a]_n$, as required.
- [Inverses] Given $[a]_n \in \mathbb{Z}_n$, we claim that $[-a]_n$ is the inverse. Indeed $[a]_n + [-a]_n = [a + (-a)]_n = [0]_n$, as required.

Having shown that the three properties of a group are satisfied, we conclude that $\mathbb{Z}_n$ is a group. ■

As a small warning, note that while our notation for abstract groups defaults to products – such as $a \cdot a \cdot a = a^3$ – in the case of groups under addition, this notation would look like $a + a + a = 3a$.

Exercise 4 Consider the subset of $\mathbb{Z}_6$ given by $G = \{[1]_6, [5]_6\}$. Determine if G is a group under multiplication modulo 6.

Exercise 5 Let X be an arbitrary set. We define the *symmetric group on X* , denoted $(S_X, \circ)$, to be the set of bijections on X with the binary operator given by function composition; namely, $S_X = \{f : X \rightarrow X, f \text{ a bijection}\}$. Show that $(S_X, \circ)$ is a group.

Example 6

Show that the non-zero rational numbers $\mathbb{Q}^*$ form a group under multiplication.

Solution. Once again, we prove the three axioms:

- [Associativity] We know that multiplication of rational numbers is associative, so we don't need to do anything here.
- [Identity] Certainly the rational number 1 is the identity. Indeed, if $r \in \mathbb{Q}^*$, then $1 \cdot r = r$.
- [Inverses] Let $p/q \in \mathbb{Q}^*$ be a rational number. We claim that its inverse is q/p , and indeed $(p/q) \cdot (q/p) = (pq)/(pq) = 1$. ■

Exercise 7 What goes wrong with Example 6 if we use $\mathbb{Q}$ instead of the non-zero rationals $\mathbb{Q}^*$?

Exercise 8 Let G be the odd rational numbers. Does this form a group under addition?

Definition 9

A group $(G, \cdot)$ is said to be *abelian* if the group operation is *commutative*; namely, for every $a, b \in G$ we have $a \cdot b = b \cdot a$.

Of the examples we've explored thus far, the only non-abelian group is the symmetric group introduced in Exercise 5. For example, let $X = \{1, 2, 3\}$ and consider the bijection σ which interchanges 2 and 3, and τ which interchanges 1 and 3. Then

$$(\sigma \circ \tau)(n) = \begin{cases} 2 & x = 1 \\ 3 & x = 2 \\ 1 & x = 3 \end{cases} \quad \text{and} \quad (\tau \circ \sigma)(n) = \begin{cases} 3 & x = 1 \\ 1 & x = 2 \\ 2 & x = 3 \end{cases},$$

which are evidently different functions. There are many other examples one can give, but a relatively straightforward one is the *dihedral group*.

Definition 10

Given a positive integer $n \geq 3$, the *dihedral group of order n* , denoted D_n , is the set of symmetries of a regular n -gon under composition.

That probably doesn't make a lot of sense, so let's break it down! A regular n -gon is a shape consisting of n sides of equal length, whose interior angles are all the same. So for example, a regular 3-gon is an equilateral triangle, while a regular 4-gon is a square. Looking at D_3 (the symmetries of an equilateral triangle) we get the 6 elements shown in Figure 1. Composition is done by applying one symmetric transformation, then another.

As shown in Figure 1, instead of labelling each of the six elements of D_3 differently, we can instead represent every symmetry as a composition of two basic symmetries. So if r represents the rotation of the triangle by 120° and s is the reflection about the vertical axis, then $D_3 = \{e, r, r^2, s, sr, sr^2\}$. As alluded to earlier, this group is not abelian! Check for yourself that $sr \neq rs$.

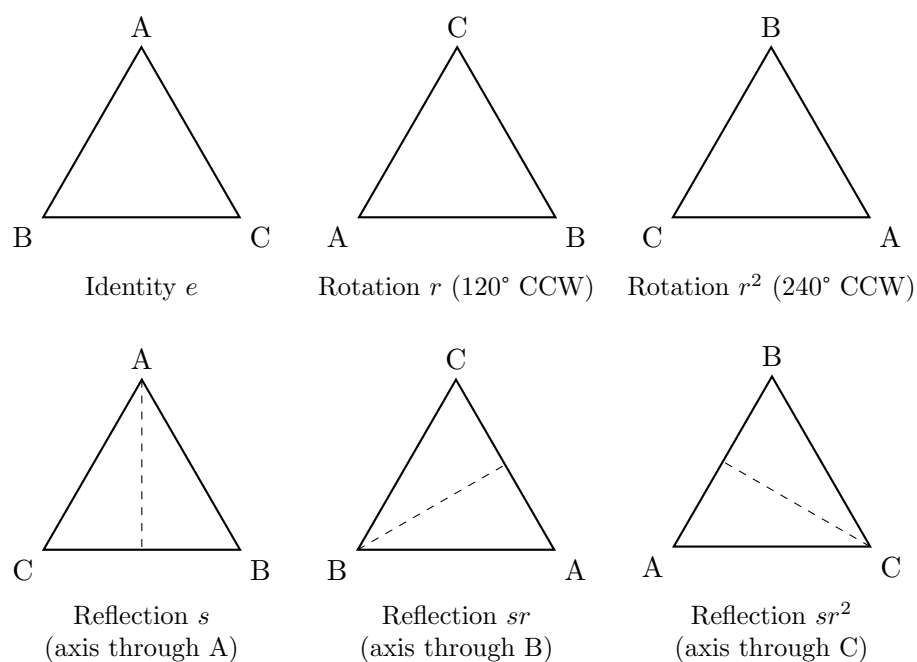


Figure 1: The 6 elements of the dihedral group D_3 .

Exercise 11 Prove that in D_3 , $sr \neq rs$. Prove this for $D_n, n \geq 3$ in general.

Exercise 12 We saw above that D_3 contains six elements. How many elements are in D_n in general?

Proposition 13

If $(G, \cdot)$ is a group, then

1. The identity element is unique.
2. If $a, b, c \in G$ and $ab = ac$, then $b = c$. Similarly, if $ba = ca$ then $b = c$.
3. The inverse of an element is unique.
4. If $a, b \in G$, then $(ab)^{-1} = b^{-1}a^{-1}$

Proof. We will give the proof of (3) and leave the others as exercises. Fix an element $a \in G$, and suppose that r and s are both inverses for a . But then $ar = e = as$. By property (2), since $ar = as$ we then have $r = s$. Thus any two inverses of a must be the same, showing that there is in fact a unique inverse. $\square$

Exercise 14 Finish the proof of Proposition 13.

Subgroups

Definition 15

Given a group G , the *order of G* is its cardinality $|G|$. Given an element $g \in G$, the *order of g* , denoted similarly as $|g|$, is the smallest $n \in \mathbb{Z}^+$ such that $g^n = e$. If no such n exists, then $|g| = \infty$.

For example, consider the group $G = \{[1]_8, [3]_8, [5]_8, [7]_8\}$ under multiplication modulo 8. Evidently we have $|G| = 4$, and

$$|[1]_8| = 1, \quad |[3]_8| = 2, \quad |[5]_8| = 2, \quad |[7]_8| = 2.$$

As another example, given D_3 we have $|D_3| = 6$, while

$$|e| = 1, \quad |r| = 3, \quad |r^2| = 3, \quad |s| = 2, \quad |sr| = 2, \quad |sr^2| = 2.$$

For an example of elements with infinite order, it's easy to check that $(\mathbb{Z}, +)$ is a group. Here the identity element is $e = 0$, which has order 1. Every other element in this group has infinite order, since as you continuously add a number to itself, it will be impossible to ever return to 1.

Proposition 16

If G is a group and $g \in G$ has order n , then $g^k = e$ for some $k \in \mathbb{Z}^+$ if and only if $n \mid k$.

Proof. ($\Leftarrow$) Suppose that $n \mid k$ and write $ns = k$ for some $s \in \mathbb{Z}^+$. Thus $g^k = g^{ns} = (g^n)^s = e^s = e$.

($\Rightarrow$) Note to begin that since $|g| = n$, then for $g^\ell = e$ to be true for some ℓ , it must be the case that $\ell \geq n$ (n is the smallest such number that gives the identity). Suppose then that $g^k = e$ for some $k \in \mathbb{Z}^+$, so that $k \geq n$. Applying the Division Algorithm, we can write $k = nq + r$ for some appropriate $q, r \in \mathbb{Z}^+$ satisfying $0 \leq r < n$. Thus

$$e = g^k = g^{nq+r} = (g^n)^q g^r = e^q g^r = g^r.$$

Since $0 \leq r < n$, it must be true that $r = 0$. Thus $k = nq$, showing that $n \mid k$ as required. $\square$

Proposition 17

If G is a group and $g \in G$ has order $n \in \mathbb{Z}^+$, then for any $k \in \mathbb{Z}^+$ we have

$$|g^k| = \frac{n}{\gcd(n, k)}.$$

Exercise 18 Prove Proposition 17. *Note:* This is a tricky proof. Just try your best.

An important concept in abstract algebra is the study of *sub-objects*. For example, you will learn about *subspaces* in your linear algebra course. Here we'll learn about subgroups.

Definition 19

If $(G, \cdot)$ is a group and $H \subseteq G$, we say that H is a *subgroup* of G if $(H, \cdot)$ is a group. In this case, we write $H \leq G$. If $H \neq G$, we instead write $H < G$.

For example, under addition we have that $\mathbb{Z} \leq \mathbb{Q} \leq \mathbb{R}$. However, it's important to keep in mind that the group operations must be the same! For example, we know that $\mathbb{Z}_6$ is a group under *addition*, and in Exercise 4 you showed that $G = \{[1]_6, [5]_6\} \subseteq \mathbb{Z}_6$ is a group under *multiplication* modulo 6. However, G is **not** a subgroup of $\mathbb{Z}_6$, since these have different binary operators. A set that is a subgroup of $\mathbb{Z}_6$ could be $H = \{[0]_6, [2]_6, [4]_6\}$ under addition.

Proposition 20: Subgroup Test

Suppose that G is a group and $H \subseteq G$ is non-empty. If for all $a, b \in H$ we have $ab^{-1} \in H$, then $H \leq G$ (under the same binary operation).

Proof. We need to show that H is a group under the same operation as G .

1. [Associativity] Since we already know that the group operation on G is associative, and $H \subseteq G$, it follows that the operation is associative on H as well.
2. [Identity] It suffices to show that the identity element $e \in G$ is also in H (why?). Since H is non-empty, choose some $h \in H$. From the hypothesis, let $a = b = h$ so that $e = aa^{-1} \in H$, as required.
3. [Inverse] Finally, let's show inverses exist in H . Fix some $h \in H$ and set $a = e, b = h$, so that the hypothesis tells us that $h^{-1} = eh^{-1} \in H$, showing that H contains h^{-1} . $\square$

Example 21

Let G be a group and fix some element $g \in G$. Define the set $H = \{h \in G : hgh^{-1} = g\}$. Show that $H \leq G$.

Solution. We'll use Proposition 20. Fix elements $a, b \in H$, so that by definition we know $aga^{-1} = g$ and $bgb^{-1} = g$. We want to show that $ab^{-1} \in G$; namely, we want to show that $(ab^{-1})g(ab^{-1})^{-1} = g$. Starting with the left hand side, we get

$$\begin{aligned} (ab^{-1})g(ab^{-1})^{-1} &= a(b^{-1}gb)a^{-1} && \text{since } (b^{-1})^{-1} = b \\ &= aga^{-1} && \text{since } b \in H \\ &= g && \text{since } a \in H. \end{aligned}$$

This is exactly what we wanted to show, so $H \leq G$. $\blacksquare$

Exercise 22 Let G be a group and define $H = \{h \in G : h^2 = e\}$. Show that $H \leq G$.

Cyclic Groups

A very special type of group are those of cyclic groups, which can be “generated” by a single element.

Definition 23

A group G is *cyclic* if there exists a $g \in G$ such that $G = \{g^n : n \in \mathbb{Z}\}$. In this case, g is said to *generate* G . If G is a (not necessarily cyclic) group and $g \in G$, then the *cyclic subgroup generated by g* is $\langle g \rangle = \{g^n : n \in \mathbb{Z}\}$.

For example, consider the group $\mathbb{Z}_4$ under addition. This group is cyclic and generated by $[1]_4$; namely, $\mathbb{Z}_4 = \langle [1]_4 \rangle$. To see this, note that the elements generated by $[1]_4$ are

$$[0]_4, \quad [1]_4, \quad [1]_4 + [1]_4 = [2]_4, \quad [1]_4 + [1]_4 + [1]_4 = [3]_4.$$

But $[3]_4$ is also a generator! Indeed, the elements generated by $[3]_4$ are

$$[0]_4, \quad [3]_4, \quad [3]_4 + [3]_4 = [2]_4, \quad [3]_4 + [3]_4 + [3]_4 = [1]_4;$$

showing that $\mathbb{Z}_4 = \langle [3]_4 \rangle$. If we look at the cyclic subgroup generated by $[2]_4$ though, we get a proper subgroup $G = \{[0]_4, [2]_4\}$.

Exercise 24 Consider the integers $\mathbb{Z}$ under addition. Is $\mathbb{Z}$ cyclic? Describe the cyclic subgroup $\langle 2 \rangle$. Generalize this to $\langle n \rangle$ for any non-zero integer.

Exercise 25 Show that a cyclic group is necessarily abelian.

Before going any further, we should verify that a cyclic subgroup is actually a subgroup.

Proposition 26

If G is a group, then for any $g \in G$ the set $\langle g \rangle$ is a subgroup of G .

Proof. This is a quick application of the Subgroup Test. Let $a, b \in \langle g \rangle$ and write $a = g^k$ and $b = g^\ell$ for some $k, \ell \in \mathbb{Z}$. Then $ab^{-1} = g^k g^{-\ell} = g^{k-\ell} \in \langle g \rangle$, since $k - \ell \in \mathbb{Z}$. Thus $\langle g \rangle \leq G$ as required. $\square$

Exercise 27 Suppose that G is a group and $g \in G$ has order n . Show that $|\langle g \rangle| = |g|$.

Exercises

1. Determine if each of the following form a group:
 - (a) $\mathbb{Z}^+$ under exponentiation.
 - (b) $\mathbb{Z}$ under the operation $a \otimes b = a + b + ab$.
 - (c) $\mathbb{Q} \setminus \{0\}$ under the operation $a \otimes b = a/b$.
 - (d) The game Rock-Paper-Scissors, denoted with the elements $\{R, P, S\}$, and a binary operator $+$ defined by the winner of the game:

$$R + P = P, \quad R + S = R, \quad P + S = S.$$

Assume that operator is commutative, so that $R + P = P + R$, etc.

2. An element of $[a]_n \in \mathbb{Z}_n$ is said to be a *unit* if $[a]_n$ has a multiplicative inverse; namely, if there exists a $[b]_n \in \mathbb{Z}_n$ such that $[a]_n \cdot [b]_n = [1]_n$. Let $\mathbb{Z}_n^*$ denote the units of $\mathbb{Z}_n$.
 - (a) Show that $\mathbb{Z}_n^*$ is a group under multiplication modulo n .
 - (b) Write out the elements of $\mathbb{Z}_{12}^*$.
3. Show that a group G is abelian if and only if $(ab)^{-1} = a^{-1}b^{-1}$ for all $a, b \in G$.
4. Let G be a group. Show that if $a_1, \dots, a_n \in G$, then $(a_1 \cdots a_n)^{-1} = a_n^{-1}a_{n-1}^{-1} \cdots a_2^{-1}a_1^{-1}$.
5. If G is a group, prove that for all $a, b \in G$ we have $|ab| = |ba|$. *Hint:* It's easier to prove that $|aba^{-1}| = |b|$, from which you can conclude the result.
6. Let $n \in \mathbb{Z}^+$ with $n \geq 3$. Show that in D_n we always have $srs = r^{-1}$.
7. If $n \in \mathbb{Z}^+$, we let S_n denote the symmetric group on $\{1, 2, \dots, n\}$. A more convenient way of writing the elements of S_n are in terms of *cycles*. A cycle is a list of numbers $(a_1, a_2, \dots, a_n)$ which describes how these numbers are mapped: $a_1 \mapsto a_2, a_2 \mapsto a_3, \dots, a_{n-2} \mapsto a_{n-1}, a_{n-1} \mapsto a_n$. Any number not mentioned maps to itself. Any permutation can be written in terms of cycles. So for example, the cycle $(1\ 2\ 6)(4\ 7) \in S_7$ describes the map

$$1 \mapsto 2, 2 \mapsto 6, 6 \mapsto 1, \quad 4 \mapsto 7, 7 \mapsto 4, \quad 3 \mapsto 3, \quad 5 \mapsto 5.$$

Like functions, cycle composition is read right-to-left, so $(1\ 2\ 3)(2\ 3) = (1\ 2)$.

Compute the corresponding cycles in S_5 :

- (a) $(1\ 3\ 4)(2\ 3)$
 - (b) $(1\ 2)(3\ 4\ 5)(1\ 5)$
 - (c) $(1\ 4\ 5)(2\ 5)(1\ 2)$
8. Show that S_n is abelian if and only if $n \in \{1, 2\}$.
9. Suppose that $\sigma \in S_n$ is a product of disjoint cycles (none of its cycles share a common number). Prove that $|\sigma|$ is the least common multiple of the lengths of its cycles.
10. If G is a group, define $Z(G) = \{a \in G : ab = ba, \forall b \in G\}$. Show that $Z(G) \leq G$.

11. If G is a group and $H \leq G$, define $C(H) = \{g \in G : gh = hg, \forall h \in H\}$. Show that $C(H) \leq G$.

12. Suppose that G is a group and $H \subseteq G$. Show that if

- For all $a, b \in H$ we have $ab \in H$, and
- For all $c \in H$, $c^{-1} \in H$,

then $H \leq G$.

13. Show that $\mathbb{Z}_8^*$ is not a cyclic group, but $\mathbb{Z}_{10}^*$ is cyclic.

14. Let G be a group and $a \in G$ have order $|a| = n$. For any $k \in \mathbb{Z}^+$, show that $\langle a^k \rangle = \langle a^{\gcd(k,n)} \rangle$.

15. If G is a cyclic group of order n , show that for any $a \in G$ we have $|a| \mid n$.

16. Show that every subgroup of a cyclic group is cyclic. *Hint:* The Division Algorithm is useful for this problem.